

WD003 Week 3 — English Worksheet

Topic: Design System — A Consistent Feel · **Course:** Portfolio Site · **Time:** about 45 minutes

This week you give your site a **design system**. You decide your colors, font, and spacing *once* — as **CSS variables** — and reuse them everywhere. When everything pulls from the same set of values, the whole site feels like it belongs together.

A CSS variable is defined in `:root` with two dashes, like `--primary: #d4380d;`. You use it anywhere with `var(--primary)`. Change it in one place, and it updates everywhere.

1 · Predict 🧙

Read each block of CSS. Before you imagine the page, write what you think will happen.

```
:root {  
  --primary: #d4380d;  
}  
h1 { color: var(--primary); }  
h2 { color: var(--primary); }
```

Both headings pull from `--primary`. If you change `--primary` to blue, how many headings change color?


```
:root {  
  --space-2: 16px;  
  --space-4: 32px;  
}  
section { padding: var(--space-4); }
```

What does `padding: var(--space-4)` do to the space inside each section? How much space is it?


```
:root {  
  --text: #333333;  
}  
body { color: var(--text); }  
h1 { color: var(--primary); }
```

`--primary` was never defined in `:root`. What color will the `h1` be?

2 · Spot the Bug 🐛

Each block of CSS below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — The body text is supposed to use the `--text` color variable, but the page shows plain black instead.

```
:root {  
  --text: #333333;  
}  
body { color: --text; }
```

Hint: a variable name on its own is not enough — it needs to be wrapped.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — This is supposed to define a primary color variable, but the value is missing its proper form.

```
:root {  
  primary: d4380d;  
}  
h1 { color: var(--primary); }
```

Hint: a CSS variable name starts with two dashes, and a hex color starts with `#`.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — Every section is supposed to have the same inside spacing using the spacing scale. Right now every section uses a different random number, so the page looks uneven.

```
.hero    { padding: 30px; }  
.about   { padding: 18px; }  
.projects{ padding: 25px; }
```

Hint: define one spacing variable and use it for all three.

Write the fixed code:

Why does using one spacing variable make the page look more consistent?

3 · Build Your Design System 📷

Open your site from last week and give it a design system in a `<style>` block.

1. In `:root`, define **three colors**: `--primary`, `--secondary`, and `--text`. Pick a palette that feels like *you*.
2. Define a **spacing scale**: `--space-1` through `--space-4` (try 8px, 16px, 24px, 32px).
3. Apply your font to the `body` (a Google Font, or a clean default).
4. Use `var(--space-4)` for the padding on every `<section>` so they all match.

When you finish, send a **photo or video** of your site with its new colors and spacing, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

The colors I chose for my variables are ...

I picked these colors because ...

I used `var(--space-4)` to ...

Before the design system, my page looked ...

After applying it, my page looked ...

If I had more time, I would change ...

4 · Record Your Walkthrough 🎥

Now take a video on your phone (or a parent's phone) while you show your styled site. Talk through it like you are teaching someone who has never used CSS variables. Try to use these words: **variable**, **color**, **spacing**, **consistent**, **reuse**.

1. Show your `:root` block and read out your color variables.
2. Change one variable's value live and show the page update everywhere.
3. Point to two sections and show their spacing matching.
4. Say why defining values once is better than repeating numbers.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.